

Hw04 IP

- Chapter 14 Problem 56

How difficult is it to expand the Great Threads fixed-cost model to accommodate another type of clothing? Assume the company produces sweatshirts. Rental cost is \$1100, variable cost per unit and selling price are \$15 and \$45, respectively, and each sweatshirt requires one labor hour and 3.5 square yards of cloth.

- [Optional] Chapter 14 Problem 65

The models in this section are called *combinatorial models*, because each solution is a combination of various 0's and 1's, and there are only a finite number of such combinations. For the capital budgeting model, there are 7 investments, so there are $2^7 = 128$ possible solutions (some of which are infeasible). Solve the model without Solver by listing all 128 solutions. For each, calculate the total cash requirement and total NPV for the model. Then, manually choose the one that stays within the budget and has the largest NPV.

Homework

Uncapacitated lot sizing (ULS)

Revisit the multi-period production example from last week.

We now extend the model to account for fixed setup costs.

- A fixed setup cost of 15,000 USD is incurred in any period in which production occurs.
- Assume that there are no capacity constraints (neither production nor storage)

$$\begin{aligned} \text{Min } & \sum_{t=1}^n (p_t x_t + h_t I_t + f_t y_t) \\ & I_{t-1} + x_t - d_t = I_t, \quad t = 1, 2, \dots, n \\ & I_0 = 5000 \\ & x_t \leq M y_t, \quad t = 1, 2, \dots, n \\ & I_t \geq 0, \quad t = 1, 2, \dots, n \\ & x_t \geq 0, y_t \in \{0, 1\} \quad t = 1, 2, \dots, n \end{aligned}$$

- Use $M \geq \sum_t d_t = 125,000$
- When solving the model in Excel, set integer optimality tolerance to 0%.
- Report the optimal production plan and setup decisions.

Multiperiod production model						
BigM	M= 125,000					
Input data						
Initial inventory	5000					
Holding cost as % of prod cost	5%					
Fixed setup cost	15,000					
Month	1	2	3	4	5	6
Production cost/unit	\$12.50	\$12.55	\$12.70	\$12.80	\$12.85	\$12.95
Holding cost/unit	\$0.63	\$0.63	\$0.64	\$0.64	\$0.64	\$0.65
Production plan (all quantities are in 100s of footballs)						
Month	1	2	3	4	5	6
Units produced						
Binary variable (produce?)						

Options

All Methods | GRG Nonlinear | Evolutionary

Constraint Precision: 0.000001

☒ Use Automatic Scaling

☐ Show Iteration Results

Solving with Integer Constraints

☐ Ignore Integer Constraints

Integer Optimality (%): 0

If the integer optimality (%) is set to or 1% or 5%, then you might not get an optimal solution. But, it may find a solution faster or within a pre-specified computational time.

Here, for this homework, set to 0.